Statement of Work

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SOW 001 for Agreement to Provide Server Migration Services to Cyberdyne Systems Corporation

Date	Services Performed By:	Services Performed For:
April 20, 2024	Illinois Tech 10 W 35th St Chicago, IL 60616	Cyberdyne Systems Corporation 123 Tech Way Los Angeles, CA 90001

Introduction

This Statement of Work (SOW) proposes two potential solutions for migrating Cyberdyne Systems Corporation's existing physical server infrastructure to a modern virtual environment that meets their requirements for security, redundancy, scalability, and future growth potential. Option 1 is a comprehensive VMware vSphere-based solution consisting of two vSphere 7 clusters built on enterprise-grade Supermicro server hardware, with shared storage on a Dell PowerVault ME484 array, providing a highly available and resilient architecture across segmented networks leveraging advanced VMware features like vMotion, HA, DRS, and vSphere Distributed Switches. Option 2 is an alternative Microsoft Hyper-V solution utilizing Windows Server 2022 Datacenter deployed as a Failover Cluster across Dell PowerEdge hosts with locally attached storage, provisioned through software-defined storage capabilities and enabled with network virtualization via VMware NSX Data Center, providing a capable virtualization platform at a lower upfront cost albeit with reduced scalability compared to VMware. Both options accomplish Cyberdyne's goal of migrating from aging physical servers to a consolidated, virtualized infrastructure for long-term growth, with the VMware solution being more robust yet costlier, while the Microsoft solution is optimized yet more limited in advanced capabilities.

Core Components

Option 1 (VMware vSphere):

The VMware vSphere 7 Enterprise Plus solution utilizes the latest version of VMware's industry-leading virtualization platform. It includes advanced features such as vMotion, Distributed Resource Scheduler (DRS), vSphere High Availability (HA), and vSphere Distributed Switch.

The VMware ESXi 7 hypervisors would run on Supermicro's SYS-620U-TNR server models, which provide a robust and scalable hardware foundation. Each server is configured with:

- 2 x Intel Xeon Gold 6346 CPUs (16 cores / 32 threads)
- 768GB of DDR4 ECC RAM
- Redundant hot-swap power supplies and fans
- RAID protected storage (4 x 960GB SSD, 4 x 8TB HDD)
- Dual 25GbE network ports

This high core count, memory capacity, and fast local storage would allow each ESXi host to run many virtual machines while still reserving resources for high availability needs.

The virtual infrastructure networks (DMZ, Production, etc.) would utilize VMware vSphere Distributed Switches to centralize network provisioning from a single interface.

Option 2 (Microsoft Hyper-V):

The Microsoft solution is based on the Hyper-V 2022 Datacenter virtualization role in Windows Server 2022. While lacking some of VMware's advanced capabilities, Hyper-V provides a robust virtualization platform at a lower upfront licensing cost.

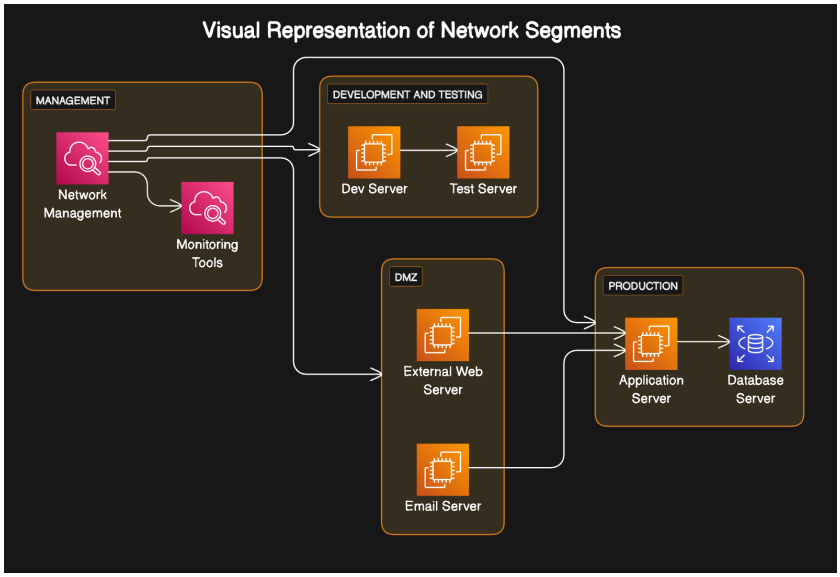
The Hyper-V hosts would run on Dell PowerEdge R750 servers, configured with:

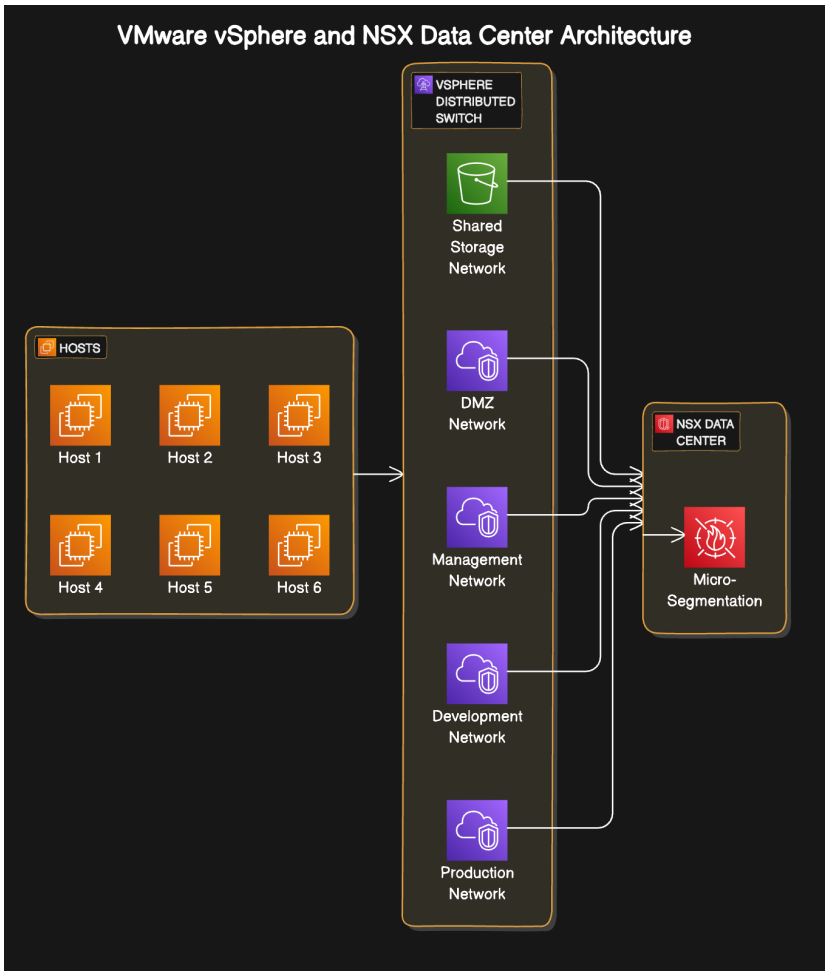
- 2 x Intel Xeon Gold 6330 CPUs (28 cores / 56 threads)
- 512GB of DDR4 ECC RAM
- Redundant hot-swap power supplies and fans
- RAID 10 storage (8 x 1.92TB SSD, 6 x 8TB HDD)
- Quad 25GbE network ports

With a higher per-socket core count, the Dell servers can host a similar number of VMs to the Supermicro solution, albeit with less memory per host.

Network virtualization would be provided by VMware NSX Data Center, allowing for streamlined provisioning of networks and micro-segmentation across the Hyper-V hosts.

Infrastructure Diagram





Redundancy and Resiliency

Option 1 (VMware vSphere):

The VMware solution implements robust redundancy and resiliency at multiple levels. This allows for high availability within each cluster, ensuring virtual machines can automatically restart on another host if one host fails.

Additionally, VMware vSphere vMotion enables live migration of running virtual machines between hosts in the same cluster with no downtime. If a host needs to be brought down for maintenance, its VM workloads can be seamlessly transitioned to other hosts.

The VMware vSphere Distributed Resource Scheduler (DRS) provides initial load balancing by intelligently placing VMs across hosts based on resource requirements. It can also dynamically migrate VMs to rebalance resources if a host becomes overloaded.

Storage redundancy is achieved by configuring the Dell PowerVault ME484 array with RAID 6 protection. Two disk drives can fail simultaneously per disk group without impacting data availability.

Network redundancy involves provisioning redundant VMkernel ports on the ESXi hosts, utilizing separate network paths to the core switching infrastructure.

Option 2 (Microsoft Hyper-V):

With the Hyper-V solution, the first level of redundancy is the Windows Server Failover Cluster configuration, which provides automatic failover of virtual machines if one cluster node fails.

Microsoft's Live Migration technology, like vMotion, allows moving running VMs between cluster nodes with minimal downtime for load balancing or maintenance operations.

Storage redundancy comes from configuring RAID 10 disk groups on the locally attached storage of each Hyper-V node. This protects against multiple simultaneous disk failures per node.

Network redundancy relies on the Hyper-V virtual switch using Hyper-V Network Virtualization with the VMware NSX Data Center overlay to provide redundant network paths.

The primary difference is that Option 1 has redundancy built-in at the virtualization cluster level with VMware vSphere, while Option 2 relies more heavily on the redundancy capabilities of the underlying Windows Server Failover Cluster and its individual nodes.

Option 1 provides a higher level of resiliency by isolating workload clusters, but Option 2 still delivers solid redundancy by stretching VM workloads across the Failover Cluster nodes. Both options utilize enterprise-grade redundant components and architectures to minimize single points of failure.

Cost Estimate

Option 1 (VMware vSphere):

Hardware	Quantity	Cost
Supermicro SYS-620U-TNR servers	13	$299,819
Dell PowerVault ME484 Storage Array (48TB raw)	1	$25,000
48-port 25GbE switches	2	$20,000
24-port 100GbE QSFP28 switches	2	$30,000
Redundant OM4 multimode fiber cabling	1	$5,000

Software	Quantity	Cost
VMware vSphere 7 Enterprise Plus	13 CPUs	$21,667/year
Veeam Backup & Replication Enterprise	13 Sockets	$8,667/year
VMware Production Support	-	$6,000/year

Total Estimated Cost: $437,486 (Year 1) **Recurring:** $36,334/year

Option 2 (Microsoft Hyper-V):

Hardware	Quantity	Cost
Dell PowerEdge R750 servers	8	$125,096
Dell PowerVault ME484 Storage Array (48TB raw)	1	$25,000
48-port 25GbE switches	2	$20,000
24-port 100GbE QSFP28 switches	2	$30,000
Redundant OM4 multimode fiber cabling	1	$5,000

Software	Quantity	Cost
Windows Server 2022 Datacenter	8 x 28-core packs	$27,992
Veeam Backup & Replication Enterprise	16 Sockets	$10,400
VMware NSX Data Center	16 CPUs	$21,120/year

Total Estimated Cost: $ 369,608 (Year 1) **Recurring:** $31,520/year

Solution Justification

Option 1 (VMware vSphere):

The VMware vSphere solution leverages VMware's market-leading virtualization platform that is widely deployed across enterprises. It meets Cyberdyne's requirements for security through network micro-segmentation, redundancy with failover clustering, and scalability to accommodate future growth.

Segmenting the virtual environment into isolated networks (DMZ, Production, etc.) via vSphere Distributed Switches provides strong multi-tenancy. Sensitive workloads like customer data can be completely separated from other systems.

The multi-cluster architecture with vMotion live migration ensures workloads remain available even during host maintenance or failures. High Availability restarts impacted VMs on other hosts.

vSphere is an extremely scalable platform. The cluster design can easily scale out by adding more ESXi host servers as needed. The vCenter centralized management simplifies operations across the entire virtual infrastructure.

Option 2 (Microsoft Hyper-V):

While not as feature rich as VMware, the Microsoft Hyper-V solution provides capable virtualization built into the Windows Server 2022 Datacenter platform Cyberdyne is already familiar with. It meets the core requirements for security, redundancy, and growth potential.

Using VMware NSX Data Center for network virtualization overlay allows provisioning micro-segmented networks and security policies for workload isolation.

The Windows Server Failover Cluster provides redundancy at the host level, allowing virtual machines to failover to other nodes if a host goes down. Live Migration supports moving VMs without downtime.

For growth, additional Hyper-V hosts can be added to the Failover Cluster, scaling out compute resources. The Hyper-V software-defined capabilities reduce management complexity as the infrastructure expands.

Growth Accommodation

Option 1 (VMware vSphere):

The VMware solution is designed to seamlessly accommodate Cyberdyne's projected 10% annual growth in storage and VM requirements. Each vSphere cluster can scale out by adding more host servers, increasing compute and memory resources.

The Dell PowerVault ME484 supports adding expansion disk shelves, allowing for cost-effective storage capacity growth. With VMware Virtual Volumes (vVols), new storage can be brought online and provisioned to VMs with minimal overhead.

VMware's CPU-based licensing model makes it simple to increase the number of CPU cores to meet growing demands without incurring significant new licensing costs.

Option 2 (Microsoft Hyper-V):

For the Hyper-V solution, growth is accommodated by scaling out the Windows Server Failover Cluster with more physical nodes. The modular architecture allows adding hosts to increase compute capacity incrementally.

Cluster Shared Volumes (CSVs) facilitate easily expanding storage by provisioning new volumes and making them accessible across all cluster nodes.

Microsoft's licensing is based on a per-core model, so new host servers can be added while only licensing the additional CPU cores brought online.

With both options, the proposed solutions are designed for scalability to meet Cyberdyne's future growth needs over the typical 3–5-year hardware refresh cycle. Growth can be managed in a cost-effective manner with simple scaling processes.

Ultimate Recommendation

Considering the requirements for security, redundancy, scalability, and future growth potential outlined by Cyberdyne Systems Corporation, the recommended solution is **Option 1 - the VMware vSphere 7 Enterprise Plus** virtualization platform.

The VMware vSphere solution provides a comprehensive, enterprise-grade virtualization architecture that meets and exceeds Cyberdyne's needs. With advanced capabilities like vMotion, High Availability, Distributed Resource Scheduler, and vSphere Distributed Switches, it delivers robust resiliency, workload balancing, and multi-tenant security through network micro-segmentation.

While the upfront costs are higher compared to the Microsoft Hyper-V alternative, the VMware solution offers superior scalability and operational efficiency that will benefit Cyberdyne over the long term. The ability to seamlessly scale compute and storage resources by adding more hosts and disk shelves simplifies future growth. Additionally, VMware's centralized vCenter management platform provides a significant advantage in oversight and administration of the large-scale virtual infrastructure.

Ultimately, the VMware vSphere architecture provides an enterprise-proven, feature-rich virtualization platform designed for the level of performance, availability, and flexibility that Cyberdyne requires for their mission-critical

workloads and future business expansion. The initial investment will be offset by lower operational costs and accelerated ROI enabled by VMware's advanced capabilities and optimization of hardware resources.

Potential Cost Savings

Compared to Cyberdyne's existing physical server infrastructure, the proposed virtualized solution offers significant cost savings in several areas:

- Reduced hardware costs: Consolidating multiple workloads onto fewer, more powerful hosts
- Lower energy and cooling costs: Fewer physical servers result in lower power consumption
- Improved operational efficiency: Centralized management and automation capabilities
- Increased hardware utilization: Ability to leverage overcommitment and dynamic resource allocation

While there are upfront costs for the virtualization software and shared storage, the long-term savings in hardware, energy, and operational costs can offset these expenses over time.

Assumptions

- The existing physical servers have the specifications outlined in the RFP (CPU cores, memory, storage).
- Cyberdyne has sufficient network infrastructure and bandwidth to support the proposed virtual environment.
- Cyberdyne will provide necessary infrastructure details (e.g., IP addressing, VLAN configurations) during the implementation phase.
- Cyberdyne will handle any required application remediation or compatibility testing for the virtual environment.

Project Timeline

The following is a high-level timeline for the migration project:

- Week 1-2: Project kickoff, detailed design, and procurement
- Week 3-4: Infrastructure setup (servers, storage, networking)
- Week 5-6: Hypervisor deployment and configuration
- Week 7-8: Virtual machine creation and migration (batch 1)
- Week 9-10: Virtual machine creation and migration (batch 2)
- Week 11-12: Testing, knowledge transfer, and project closure